TECHNICAL SUPPLEMENT 02 · ENGLISH

Method, Scenario and Interaction Analysis

Agri-Social Navigation in Vineyard Corridors

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Study identity	Specification
Campaign / Campaña	EVAL_20261019_112257
Navigation / Navegación	9 × 4 × 10 = 360; SUCCESS 266; TIMEOUT 94
Benchmark	180; 20 per method / por método
Version / Versión	20 October / octubre 2026

Numerical companion to the revised Sensors manuscript. This report uses the identified endpoint dataset and separates it from the older public campaigns. All seven authors are listed in the original manuscript order.

1. Model and assumptions

Working model: Y = intercept + Method + Scenario + Method×Scenario + error, with sum coding and independent HC3-robust inference. Matched seed trajectories are not verified, so repeated-measures ANOVA, Greenhouse-Geisser corrections and paired random-intercept inference are not presented as established features of this dataset.

Exposure12_s

Quantity	Recorded value
n	360
cells	36
expectedCells	36
designRank	36
covarianceRank	35
parameterCount	36
status	ESTIMATED_WITH_DIAGNOSTICS
shapiroResidual_p	2.631766428108771e-32
breuschPagan_p	0.07563102780151795
brownForsythe_p	2.3790396592152652e-07

Term	F / statistic	p	df numerator	df denominator
Intercept	14972.857	<1e-15	1	324.000
C(Method, Sum)	66783.303	<1e-15	8	324.000
C(Scenario, Sum)	19.044	2.126e-11	3	324.000
C(Method, Sum):C(Scenario, Sum)	1024.071	<1e-15	24	324.000

DurationCompleted_s

Quantity	Recorded value
n	266
cells	28
expectedCells	36
designRank	--
covarianceRank	--
parameterCount	--
status	NOT_ESTIMABLE_FULL_FACTORIAL_MISSING_CELLS
shapiroResidual_p	--
breuschPagan_p	--
brownForsythe_p	--

Failure

Quantity	Recorded value
n	360

Quantity	Recorded value
cells	36
expectedCells	36
designRank	36
covarianceRank	4
parameterCount	36
status	ESTIMATED_WITH_DIAGNOSTICS
shapiroResidual_p	1.913184951202863e-32
breuschPagan_p	1.1793381088445718e-33
brownForsythe_p	5.197448344612398e-29

IntimateEvent

Quantity	Recorded value
n	360
cells	36
expectedCells	36
designRank	36
covarianceRank	0
parameterCount	36
status	ESTIMATED_WITH_DIAGNOSTICS
shapiroResidual_p	None
breuschPagan_p	None
brownForsythe_p	None

2. What is estimable

Exposure is observed in all 36 cells. Its design rank is 36; the covariance rank and residual diagnostics qualify the omnibus tests. Completed duration has only 28 populated cells, preventing a full 36-cell factorial duration model. Binary linear-probability covariance is singular (failure rank 4; intimate-event rank 0); its Wald omnibus terms are not interpreted as confirmatory evidence. Exact event comparisons are reported separately.

Source and scope

Source: data/; results/expected/. PDFs provide the reading reports; CSV/JSON retain numerical inputs and provenance. Scope: evaluated simulation, without new experiments, independent raw-stream reconstruction or physical validation.